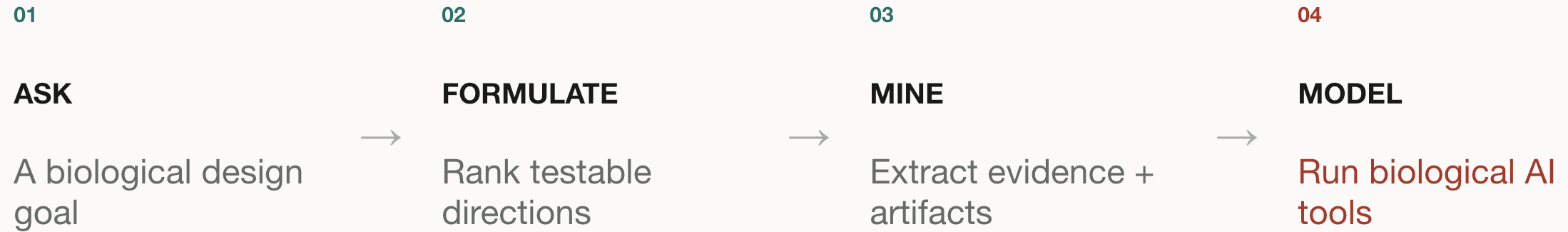


We asked AI to design an impossible protein.

It tried anyway.

A scientist-controlled agent that turns impossible biological questions into evidence, models, and candidate designs.

One scientist. Three coordinated scientific agents.



Inspect, redirect, or stop at every transition.

Formulation frames it. Literature grounds it. Proto-tools computes it.

The formulation agent turns goals into testable routes.

SCIENTIST STARTS WITH

Design an RF-responsive biological system

OPEN-ENDED

A vague biological goal

Mechanism and evaluator still unknown



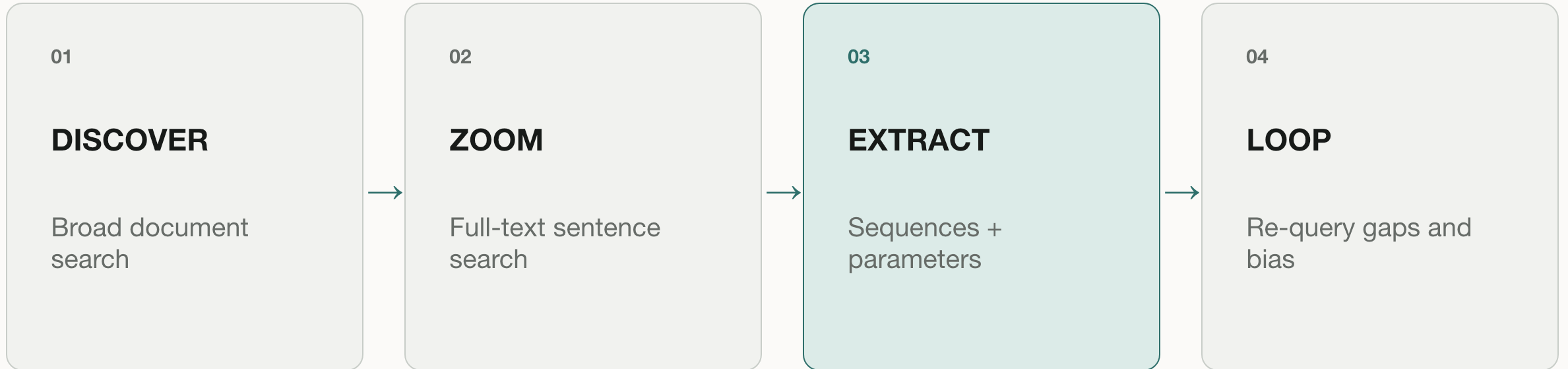
RANKED

Ranked, falsifiable research routes

Drop, follow up, or refine

Inject priors. Redirect at any time.

The literature agent turns papers into usable artifacts.



Output: traceable mechanisms, sequences, and quantitative constraints.

Proto-tools can co-fold a protein with its ligand.



CO-FOLDING DEMO

- 1 Select the right structure model
- 2 Run KRAS G12C + sotorasib
- 3 **Return structure + confidence**

ligand ipTM 0.976 · pTM 0.928

The scientist stays in control at every handoff.

01

shape the objective

Add priors + constraints

02

inspect the evidence

Open papers + artifacts

03

approve computation

Choose models + review outputs

AUDIT TRAIL

Question · queries · papers · artifacts · model jobs · results

Every handoff remains inspectable, reproducible, and reversible.

Make the impossible. Scientifically actionable.

Search, reason, and compute—with the scientist in control.

Ask. Inspect. Redirect. Design.